



Self-Supervised Models of Speech Processing for Haitian Creole

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Abstract

We develop tailored speech processing models for Haitian Creole positioning it as a high-resource language in terms of SSL models of speech processing. We do so by pretraining monolingual Wav2Vec2-Base, Wav2Vec2-Large and Data2Vec-Audio-Base models from scratch, which are then finetuned on an Automatic Speech Recognition task. We compare the performance of these models with models finetuned from larger multilingual (XLSR-53, XLSR2-300m, MMS-1B) and monolingual French-based models (LeBenchmark 1 to 7K). Our results highlight the effectiveness of pretraining monolingual models from scratch, demonstrating that they can achieve performance comparable to or even surpassing larger models derived from large-scale multilingual pretraining. Our work provides essential resources for robust Haitian Creole speech recognition and offers pretrained models that can be adapted to other French-based Caribbean Creoles, opening new avenues for linguistic research and practical applications in the region.

Index Terms: wav2vec2, data2vec, Haitian Creole, speech recognition

1. Introduction

Haitian Creole (*kreyòl*, or *kreyòl ayisyen*) is the primary language spoken in the Republic of Haiti, with approximately 11 million speakers. Being spoken in a context of diglossia, Haitian Creole (HC) still faces challenges in achieving parity with French, which is considered a more prestigious variety, despite efforts within Haiti to grant it an equal status.

In this paper, we present our work on the development of self-supervised speech processing models pretrained exclusively on HC,¹ which are further finetuned on a Automatic Speech Recognition (ASR) task. The monolingual models we train represent a departure from prior approaches for this language that exclusively relied on transfer learning from multilingual models or from a typologically close high-resource language, and more specifically here, from its lexifier language, French. Our work grants Haitian Creole an equal status to other high-resourced languages such as English [1, 2], French [3] or Mandarin Chinese [4] which all have dedicated monolingual models. In summary, our work shows that monolingual models perform competitively with similar-sized models trained via transfer learning for ASR, despite using less training data.

Beyond the technical aspect, our initiative carries significant symbolic weight. For too long, it has been asserted that Creole languages were destined to disappear in favour of their

lexifier languages (a process called ‘decreolization’, see [5] for a critical review), in this case, French. This has been theorised as the “pidgin → creole → decreolisation cycle”. Our initiative highlights that HC can receive the same attention as ‘major languages’ and that HC speakers should and *now do* have equal access to advanced language technologies, just as speakers of high-resourced languages do.

2. Background

The field of speech processing for Creole languages is relatively sparse, compared to other languages such as English or French, mainly due to the lack of accessible data. One of the very first datasets created to train speech processing models for HC, and more specifically ASR models, was collected by the DARPA DIPLOMAT project in the late 1990s [6], and is commonly known as the HAITI-CMU dataset. More recently, a larger collection of speech was collected for the IARPA BABEL program, which is accessible through the Linguistic Data Consortium [7] (see section 3 for more details on these two datasets). To the best of our knowledge, other than the two previous datasets we mentioned, no publicly available datasets of transcribed Standard Haitian Creole exist, which severely limited the work done for this language. This data scarcity is not limited to the spoken realm but also to textual approaches, and it is only recently that initiatives to develop textual resources specifically targeting Creole Languages emerged.

Over the years, most work on Haitian Creole (HC) has primarily focused on Automatic Speech Recognition (ASR) [8, 9]. However, in most work, ASR is often treated as a component of a larger speech translation pipeline [6, 10], where transcriptions serve as input for machine translation models in a cascaded approach. Beyond ASR, research on spoken HC has also explored keyword spotting [11, 12], driven by datasets such as the IARPA Babel Haitian Creole Language Pack [7], which was specifically created for this purpose. Research targeting other French-based Creoles is similarly limited. Recently, ASR models have been developed for Guadeloupean Creole, which is related to HC and mutually intelligible [13, 14, 15]. Regarding Mauritian Creole – spoken in Mauritius, which is neither related to HC nor mutually intelligible – the field appears more active, with notable contributions from [16, 17, 18] and part of the work by [13], although research remains relatively niche.

Despite recent personal initiatives to create modern speech processing models for HC using transfer learning of pretrained multilingual models like Whisper [19], XLS-R [20] or MMS [21], *reproducible scientific* efforts to develop monolingual models comparable to Wav2Vec2 for English or LeBENCHMARK for French are, to our knowledge, nonexistent. The only exception is the work of [22] which is limited in scope and do-

¹The models are made available on <https://huggingface.co/LLL-CREAM> and the *exact and exhaustive* list of speech segments used is made available on <https://gin.g-node.org/CREAM/SSL-Haitian> allowing full reproducibility.

Table 1: (1a) shows the distribution of recording durations for each split. Pre-VAD refers to the duration of the recordings before any VAD algorithm was applied, while Post-VAD indicates the duration of the recordings after retaining only the segments identified as speech by the VAD. (1b) presents the gender distribution within the datasets (M[ale]/F[emale]/UNK[nown]). The symbol “-” is used when the value is unknown. Count represents the number of unique speakers. Each dataset is referred to by its acronym, which is introduced in parentheses following its full name in section 3.

(a)									(b)				
Split	Dataset Type	BABEL	CNHC	RADIO-H	ALH	H-CMU	VOXLG	Overall Total	Dataset	Count	M	F	UNK
Train	Pre-VAD	176h 57m	5h 03m	1131h 11m	345h 36m	16h 04m	73h 47m	1748h 40m	ALH	216	117	42	57
	Post-VAD	74h 03m	4h 05m	1001h 15m	221h 55m	12h 18m	73h 47m	1387h 25m	BABEL	1091	547	544	-
Val	Pre-VAD	5h 25m	2h 03m	88h 08m	5h 16m	1h 29m	11h 32m	113h 57m	CNHC	-	-	-	-
	Post-VAD	2h 11m	1h 35m	75h 03m	3h 32m	1h 07m	11h 32m	95h 02m	H-CMU	152	80	59	13
Test	Pre-VAD	20h 02m	2h 09m	114h 36m	5h 26m	2h 03m	10h 35m	154h 53m	RADIO-H	1081	-	-	1081
	Post-VAD	8h 19m	1h 37m	100h 44m	4h 01m	1h 27m	10h 35m	126h 45m	VOXLG	7007	-	-	7007
Total	Pre-VAD	202h 26m	9h 16m	1333h 56m	356h 18m	19h 37m	95h 55m	2017h 31m	Overall Total	9547	744	645	8158
	Post-VAD	84h 34m	7h 17m	1177h 03m	229h 29m	14h 53m	95h 55m	1609h 14m					

main, as it relied on 250 hours of spoken data, which solely consisted of fieldwork recordings from a unique dataset.

In summary, there is a significant gap between the state-of-the-art speech processing models available for languages like English or French and those for HC. This gap is largely due to the lack of accessible data, which prevents the development of comparable models. Additionally, past speech processing efforts for HC, primarily ASR models, have remained largely confidential and inaccessible. Our work aims to address these challenges by (1) surveying the existing datasets of spoken HC, (2) training state-of-the-art self-supervised monolingual WAV2VEC2 and DATA2VEC models, and (3) finetuning these models on an ASR task while making them freely available to the creolist community, our primary target audience.

3. Haitian Creole Speech Corpora

This section surveys the datasets of spoken Haitian Creole used in our experiments. Table 1a offers a concise comparison of each dataset, highlighting their size, train/validation/test split partitioning, and gender distribution among speakers.

ATLAS LINGUISTIQUE D’HAÏTI (ALH) The ATLAS LINGUISTIQUE D’HAÏTI (*Linguistic Atlas of Haiti*) [23] consists of a collection of 499 audio recordings collected in Haiti between 1978 and 1987 for the purpose of creating a linguistic atlas, and totalises 356h of audio. The recordings were originally done on audio cassettes with tape recorders, and were then digitalised by the French National Library (*Bibliothèque Nationale de France, BNF*) in 2010. Each recording is on average 45 minutes long and features one or several interviewers eliciting words or phrases from their native collaborators. This data has been made *publically* and *legally* available by the BNF and the *Faculté de Linguistique Appliquée de l’Université d’État d’Haïti* and is accessible through the COCOON platform [24]. This corpus consists entirely of raw speech. Each recording is enriched with detailed metadata, including the exact location of collection and speaker information (e.g. age, profession, literacy). The recordings span the entire country, ensuring a large diversity of accents and vocabulary, by more than 200 speakers.

CORPUS OF NORTHERN HAITIAN CREOLE (CNHC) The CORPUS OF NORTHERN HAITIAN CREOLE [25] consists of 10 recorded and transcribed interviews conducted in Cap-Haïtien (Northern Haiti) to study dialectal variation in relation to standard Haitian, and totalises 9h of audio. The transcriptions use non-standard spellings to more faithfully represent speak-

ers’ pronunciations. For example, standard/non-standard variations include: “*Pòwoprens/Pòtoprens*” (Port-au-Prince, capital city of Haiti); “*eske/eskeu*” (question word); or “*de/deu*” (two). While we do not have the exact number of speakers, we estimate it to be between 10 and 20, all from the Cap-Haïtien region.

HAITI-CMU (H-CMU) The HAITI-CMU dataset consists of recordings that were originally collected for the DARPA DIPLOMAT project [6] in the late 1990s and totalises 20h of speech. The dataset solely consists of read speech, using sentences stemming from various sources (novels, speeches, religious material, etc.), resulting in more than 7.5k sentences for 152 speakers. The recordings were collected in various locations (U.S. & France) with speakers from different backgrounds and regions of Haiti. Some of the speakers having lived or having been brought up in Metropolitan France, or Overseas France islands where neighbouring Creoles (Guadeloupean, Martinican, Guianese) are spoken, this might have affected their accent. All the recordings are transcribed using the standard Haitian Creole spelling conventions. They have been collected in a noise-free environment resulting in high-quality recordings.

IARPA-BABEL HAITIAN CREOLE (BABEL) The IARPA-BABEL HAITIAN CREOLE dataset [7] consists of recordings in Haitian Creole that were gathered for the U.S. government in the course of the BABEL program. The recordings were collected in 2012/2013 and mainly consist of conversational speech recorded over the phone. Some recordings were also collected using scripted material, where speakers were asked to read specific sentences and/or discuss specific topics. The whole dataset consists of more than 11k recordings totalising more than 202h of speech across more than 1k speakers. Most recordings were transcribed using standard Haitian Creole spelling conventions, but some were left untranscribed. To ensure maximal diversity, speakers come from different region of Haiti, and span a wide age range, and is balanced between male and females. The recordings were done in various environmental conditions, from very noisy (e.g. street) to very calm environments (e.g. home or office).

VOXLINGUA HAITIAN (VOXLG) VoxLingua [26] is a speech dataset that consists of audio recordings scraped from YouTube, in order to train language identification models. The whole dataset consists of recordings for 107 languages. In this work, we only use the recordings that were classified as being in Haitian Creole, totalising 96h of speech. The distributed recordings are already preprocessed [26] in order to exclude spoken sections from noise and non-spoken sections using a

voice activity detection (VAD) model. A speaker diarisation model was also applied to separate speakers from one another, resulting in more than 36.7k segments for 664 original recordings. According to the speaker diarisation model, the dataset consists of more than 7k unique speakers. Given there are no ground-truth annotations, this is only an estimate and should be interpreted with caution, as this number seems to largely overestimate the true number of speakers.

RADIO HAITI (RADIO HAITI) While the datasets previously mentioned are known to the speech and language processing community, and have already been used in other publications, RADIO HAITI is a new dataset, which is used for the first time in this publication. RADIO HAITI consist of recordings aired by *Radio Haiti-Inter* between 1957 and 2003, which were made available by Duke University through the *David M. Rubenstein Rare Book Manuscript Library* [27]. The dataset comprises over 5,000 recordings totaling 2,400 hours of spoken content. About half of the recordings only consist of speech in Haitian Creole, while the other half is mostly in French or a mix of French and Haitian Creole, and, to a rarer extent English or Spanish. In this work, we only use a subset of this dataset, specifically the portion in Haitian Creole. This portion consists of about 2.9k recordings totalling more than 1.3k hours. This is, to the best of our knowledge, the largest ever collection of speech in Haitian Creole. Although most recordings were made in a studio, ensuring noise-free, high-quality audio, the dataset also includes interviews and interventions conducted over the phone or in public spaces, introducing a wide range of recording conditions. Each recording is paired with metadata that specifies the main topics discussed and the key speakers, typically the presenters and, when relevant, the interviewees. Although it is impossible to determine the exact number of speakers (e.g. speakers who intervene over the phone or during interviews are not mentioned in the metadata), the recordings we use feature *at least* 1,000 unique individuals. Similarly, while we noticed the speakers are from different backgrounds and have, for some, a distinct accent (e.g. from the Cap Haïtien region), we do not have metadata indicating where the speakers come from. Finally, the metadata specifies the airdate of the recordings (at least the year). Most of the recordings that are used in this publication were aired from the 1980s to the early 2000s.

Overall, our work uses more than 2000h of recordings in Haitian Creole. While we cannot give an exact number of speakers, we estimate it is over 3,000 unique speakers. The recordings cover a wide range of environmental conditions, from fieldwork recordings done in the field (ALH) or in calm environments (CNHC), to read (H-CMU, BABEL) speech and to more natural conversational speech (BABEL, VOXLG, RADIO-H). Concerning gender balance, although we lack this information for all the datasets we use, when it is available, we observe a fairly balanced distribution with 744 males and 645 females.

4. Pretraining

To ensure consistency across all datasets, we applied a standardised preprocessing pipeline to each dataset before training our SSL models. First, all recordings were resampled to mono PCM 16 bits, 16 kHz. A voice activity detection (VAD) model [28, 29] was then applied to each recording in order to isolate spoken sections from silences and non-human speech sounds. The resulting segments were then combined through iterative concatenation, preserving temporal order so as to reach a mean

Table 2: *Hyperparameters of the best-checkpoint models, including inner & output Transformer block dimensions, number of attention heads, and training statistics.*

Model	Output Dim.	Inner Dim.	Attn. Heads	Transf. Blocks	Train. Steps	# Train. GPUs	GPU hours
data2vec-base	768	3072	12	12	292k	16	3626.65
wav2vec2-base	768	3072	12	12	225k	8	3739.67
wav2vec2-large	1024	4096	16	24	218k	16	4340.99

duration of 20 seconds.² In the rare cases where VAD segments were larger than 30s, the segments were further segmented into sub-segments of a maximum length of 30s.³ Eventually, the concatenated segments reached a mean duration of 21.9s with a standard deviation of 7.3s. Given that VOXLG was already preprocessed in a similar fashion by [26] we did not apply our pipeline to this dataset. Overall, using a VAD preprocessing step resulted in a 25% reduction of the total amount of audio. This value largely varies from one dataset to the other, ranging from 57% for BABEL to only 11% for RADIO-H (see Table 1a).

The train/validation/test partitions were constructed so as to ensure that the validation set includes speakers not seen during training, and the test set includes speakers not seen during training or validation. This helps ensuring to select the checkpoint that generalises best on unseen speakers. To ensure maximal reproducibility and coherence with prior works, all our models were trained using PyTorch [31] through the *fairseq* toolkit [32]. The hyperparameters for the architecture of the WAV2VEC2-BASE and WAV2VEC2-LARGE models are identical to [33], and for the DATA2VEC model identical to [2].⁴ The architectures only mildly differ from those presented in [1] with for example 12 attention heads for our BASE version instead of 8. Instead of training the models for a fixed amount of steps (e.g. [3] trains their 1k BASE and LARGE models for 200k steps) we decided to manually stop training when the models had converged, where convergence is defined as the point where the training and validation curves intersect. The main hyperparameters of our models are shown in Table 2.

5. Finetuning

Experimental Settings. Once our WAV2VEC2-BASE, WAV2VEC2-LARGE and DATA2VEC-AUDIO-BASE models pretrained, we finetuned them on an ASR task. Alongside our three pretrained model, we also finetune XLSR53-56k and XLSR2-300m-large (whole model) [20] and MMS-1B-all and MMS-1B-FL102 (using adapter layers) [34]. To do so, we used the transcribed recordings of the BABEL and H-CMU datasets. Contrary to CNHC, both consist of transcriptions that follow the standard orthography devised by the National Teaching Institute [35], which is commonly used by official Haitian institutions. We re-used the same train/validation/test partitions that were used for pretraining with the only difference that we excluded recordings of the BABEL corpus that were not

²This value was chosen so as to obtain segments of similar length when accounting for standard deviation while limiting the number of segments longer than 30s. For reference, [1] and [3] use 20-second-long segments for the BASE and LARGE models, while [3] use 15-second-long segments for their XLARGE model, and [30] use 15-second and 20-second segments for their BASE and LARGE models, respectively.

³This could potentially result in a single word being split across two segments, however, the number of hard crops performed is minimal, and this method was only applied to segments from the RADIO-H corpus.

⁴DATA2VEC-AUDIO-BASE uses a WAV2VEC2-BASE backbone.

Table 3: WER and CER (mean $\pm$ std. dev.) at global level, by gender, and by dataset, sorted by global CER on the test set (best checkpoint selected based on validation results). The lowest CER is in **blue**, the lowest WER is in **red**, and the lowest mean CER/WER per category (F/M or BABEL/H-CMU) are in **bold**. Bold model names indicate finetuned versions of our pretrained models.

Model	Global				Gender								Dataset							
	CER		WER		CER				WER				CER				WER			
					F		M		F		M		BABEL		H-CMU		BABEL		H-CMU	
	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std
data2vec-hat-1.4K-base	21.0	51.4	33.7	49.9	19.3	41.0	23.0	60.2	31.9	43.4	35.7	55.8	22.9	53.4	1.5	6.1	36.4	51.3	5.0	9.5
wav2vec2-hat-1.4K-large	21.3	53.0	34.8	49.7	19.7	43.2	23.1	61.5	33.2	43.9	36.8	55.1	23.1	55.1	1.4	6.1	37.6	51.1	5.3	9.7
wav2vec2-hat-1.4K-base	21.5	52.0	34.5	52.8	19.7	41.1	23.5	61.1	32.7	44.1	36.8	60.4	23.4	54.0	1.5	6.4	37.3	54.3	4.8	9.1
xlsr2-300m-large	21.5	50.1	35.5	51.0	19.7	39.4	23.6	59.1	33.6	43.4	37.7	57.7	23.4	52.0	1.6	5.9	38.3	52.4	5.9	10.1
wav2vec2-FR-7K-large	22.5	51.7	36.8	48.6	20.8	42.0	24.5	60.0	35.3	43.7	38.7	53.1	24.5	53.6	1.5	6.0	39.8	49.7	5.3	9.7
wav2vec2-FR-3K-large	23.4	51.5	38.3	48.5	21.5	41.2	25.5	60.2	36.6	42.3	40.4	54.1	25.4	53.4	1.6	6.1	41.4	49.6	5.4	9.6
wav2vec2-FR-7K-base	23.8	52.9	38.4	50.6	22.0	42.7	25.8	61.5	37.0	45.1	40.2	55.7	25.9	54.8	1.6	6.2	41.5	51.8	5.5	9.6
wav2vec2-FR-2.6K-base	24.6	54.8	39.4	53.9	22.9	46.7	26.5	62.0	37.6	47.1	41.6	60.1	26.7	56.8	1.6	6.3	42.6	55.3	5.3	9.5
wav2vec2-FR-3K-base	24.8	55.5	40.0	54.7	23.4	47.1	26.5	63.1	38.1	45.9	42.3	62.4	27.0	57.6	1.8	6.4	43.2	56.1	6.2	9.9
wav2vec2-FR-1K-large	26.5	48.7	45.8	47.8	24.7	40.0	28.6	56.1	43.8	42.9	48.4	52.3	28.8	50.3	2.2	6.5	49.3	48.4	8.4	11.1
wav2vec2-FR-1K-base	27.0	54.8	44.2	53.4	25.1	47.3	29.1	61.6	41.9	46.0	47.0	59.9	29.3	56.7	1.9	6.0	47.7	54.4	6.9	13.1
mms-1b-all	30.7	51.3	58.2	49.0	29.3	45.5	32.3	56.6	56.4	43.8	60.4	53.8	33.2	52.9	3.6	6.9	62.1	49.3	16.1	14.9
mms-1b-fl102	31.6	49.3	61.7	46.9	30.2	44.8	33.4	53.5	59.8	43.3	64.0	50.3	34.1	50.8	5.3	6.9	65.2	47.3	24.2	15.8
xlsr53-56k	34.7	54.2	66.4	50.7	33.2	46.9	36.7	60.8	64.2	46.2	69.2	54.9	37.5	55.9	5.6	7.0	70.2	51.2	26.0	16.5

transcribed. Hence, the train/validation/test partitions for this dataset are subsets of the SSL pretraining partitions. For both datasets, all the speakers in the validation and test sets have not been seen at training time (and not seen at validation time for the test set). Contrary to pretraining, where we used concatenated segments obtained by a VAD model, we used here the gold segmentation to segment the recordings in smaller chunks. The models were trained with CTC loss and frozen the first 10k steps to avoid overfitting. We used Viterbi decoding without any external LM, with a beam of 5.

Text Preprocessing. We use a vocabulary of 28 symbols (27 characters and an additional space character $\langle \rangle$).⁵ When encountering foreign graphemes in loanwords or proper nouns (e.g. $\langle qu \rangle$ as in “Inuqua”) we converted them to their closest equivalent (e.g. $\langle k \rangle$ as in “Inuka”). Digits were fully spelled out (e.g. “21”: “venteyen”) using the `num2words` package that we adapted for Haitian Creole,⁶ and abbreviations were fully spelled out (e.g. “HTML”: “ach te èm èl”) using a custom script. All the annotations of the IARPA Babel specifications (e.g. to signal noise, hesitations, etc.) were removed. Finally, the transcriptions were lowercased, all the punctuation was removed and characters normalised using Unicode’s NFKC standard [36]. Finally, we separated clitic pronouns (“t”, “m”, “n”, “y”, “l”, and “w”) which are phonetically bound (e.g. “w” as in “papa w”, your father) from the words they attach to. Haitian Creole being primarily an oral language, there is significant variation in their writing, often using hyphens, apostrophes, spaces, or direct concatenation without strict regularity

Results & Discussion. Table 3 shows Word and Character Error Rates (WER/CER) at global level, by gender and by dataset. The best models are our monolingual models pretrained on 1.4K hours of speech in HC (with a CER of 21 and WER of 33.7 for the best of them, DATA2VEC-AUDIO-BASE), closely followed by XLSR-300m-large. Surprisingly, despite having been finetuned on the same data, XLSR53-56k and the two MMS models lie far behind. We observe that our models, despite having fewer parameters (95M for BASE models and 317M for LARGE) obtained better scores than larger models (MMS-1B), and better scores than models trained on many more recording hours and more language (1k languages for MMS, 100+ for XLSR). Prior works such as [37] found that transfer from a

closely related language (e.g. Czech to Slovak) can outperform large multilingual models like XLSR. In contrast, even our most targeted transfers – finetuning from French WAV2VEC2 models – underperform XLSR-300m-large, despite French being very close to HC.

We observe very high standard deviations across all models. Upon closer examination, this can be attributed to the substantial differences in error rates (ER) between H-CMU and BABEL. H-CMU consists of read speech, recorded in calm environments, while BABEL consist of natural speech recorded in noisy environments at 8kHz using phones. The difference lies both in the *nature* of the spoken content (read v. natural) and the intrinsic acoustic quality of the recordings, with BABEL known to be a “*challenging speech recognition benchmark*” [20]. Finally, we explored ER by gender. As largely documented [38], ER are lower for females than for males. The mean difference between male and female across all models is of 4 WER and 3.6 CER points, with large variations across models (5 WER points for the model finetuned from wav2vec2-FR-1K-base and 3.2 for wav2vec2-FR-7K). Our models lie inbetween, with the model finetuned from wav2vec2-hat-1.4K-base having a 4 WER points difference between males and females.

In summary, our monolingual HC models, pretrained on 1.4K hours of speech, outperform larger multilingual models, underscoring the value of domain-specific pretraining.

6. Conclusion

In this work, we presented three pretrained models of speech processing for Haitian Creole based on the WAV2VEC2-BASE, WAV2VEC2-LARGE, and DATA2VEC-AUDIO-BASE architectures. We further finetuned these models on a speech recognition task and show they obtain competitive results compared to large-scale multilingual and monolingual models. This is the first time models of this scale are developed for Haitian Creole, establishing it as a high-resource language in terms of SSL models of speech processing. Our work provide the necessary tools for robust speech recognition, and source models for transfer learning to other French-based Caribbean Creoles, and contribute to grant Haitian Creole equal status alongside French. We hope that creolists will seize upon these models to transcribe data on a large scale. They have already been used to transcribe 1,600 hours of RADIO HAITI recordings, enabling unprecedented study of linguistic phenomena in Haitian Creole.

⁵aàbcdeēfghijklmnoōprstuvwyz|

⁶<https://github.com/LLL-Orleans/num2words>

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8. References

- [1] A. Baevski *et al.*, “wav2vec 2.0: a framework for self-supervised learning of speech representations,” in *NeurIPS '20*. Red Hook, NY, USA: Curran Associates Inc., 2020.
- [2] —, “data2vec: A general framework for self-supervised learning in speech, vision and language,” in *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA*, ser. Proceedings of Machine Learning Research, K. Chaudhuri *et al.*, Eds., vol. 162. PMLR, 2022, pp. 1298–1312.
- [3] T. Parcollet *et al.*, “LeBenchmark 2.0: A standardized, replicable and enhanced framework for self-supervised representations of french speech,” *Computer Speech & Language*, vol. 86, p. 101622, 2024.
- [4] K.-H. Lu *et al.*, “A context-aware knowledge transferring strategy for ctc-based asr,” in *SLT2022*, 2023, pp. 60–67.
- [5] M. DeGraff, “Against creole exceptionalism,” *Language*, vol. 79, no. 2, pp. 391–410, 2003.
- [6] R. Frederking *et al.*, “Interactive speech translation in the DIPLOMAT project,” in *Spoken Language Translation*, 1997.
- [7] Andrus, Tony *et al.*, “IARPA Babel Haitian Creole Language Pack IARPA-babel201b-v0.2b,” 2017.
- [8] N. T. Vu *et al.*, “Initialization schemes for multilayer perceptron training and their impact on asr performance using multilingual data,” in *Interspeech 2012*, 2012, pp. 2586–2589.
- [9] W. Breiter, “Rapid bootstrapping of haitian creole large vocabulary continuous speech recognition,” 2014.
- [10] R. Frederking *et al.*, “Interactive speech translation in the diplomat project,” *Machine Translation*, vol. 15, no. 1/2, p. 27–42, Jun. 2000.
- [11] V.-B. Le *et al.*, “Developing STT and KWS systems using limited language resources,” in *Interspeech 2014*, 2014, pp. 2484–2488.
- [12] M. J. F. Gales *et al.*, “Speech recognition and keyword spotting for low-resource languages: Babel project research at cued,” in *SLTU*, 2014.
- [13] C. Macaire *et al.*, “Automatic speech recognition and query by example for creole languages documentation,” in *ACL 2022*, S. Muresan *et al.*, Eds., Dublin, Ireland, May 2022, pp. 2512–2520.
- [14] E. Le Ferrand *et al.*, “Outiller la documentation des langues créoles,” in *LIFT 2023*, Vandoeuvre-Lès-Nancy, France, Nov. 2023.
- [15] É. Le Ferrand *et al.*, “Automatic transcription of grammaticality judgements for language documentation,” in *COMPUTEL*, S. Moeller *et al.*, Eds. St. Julians, Malta: Association for Computational Linguistics, Mar. 2024, pp. 33–38.
- [16] W. Noormamode *et al.*, “A speech engine for mauritian creole,” in *Information Systems Design and Intelligent Applications*, S. C. Satapathy *et al.*, Eds. Singapore: Springer, 2019, pp. 389–398.
- [17] N. Gooda Sahib-Kaudeer *et al.*, “Automatic speech recognition for kreol morisien: A case study for the health domain,” in *Speech and Computer*, A. A. Salah *et al.*, Eds. Cham: Springer International Publishing, 2019, pp. 414–422.
- [18] B. Gobin-Rahimbux *et al.*, “A voice-based personal assistant for mental health in kreol morisien,” *JECE*, vol. 2023, Dec. 2023.
- [19] A. Radford *et al.*, “Robust speech recognition via large-scale weak supervision,” in *Proceedings of the 40th International Conference on Machine Learning*, ser. ICML'23. JMLR.org, 2023.
- [20] A. Babu *et al.*, “XLS-R: Self-supervised cross-lingual speech representation learning at scale,” in *Interspeech 2022*, 2022, pp. 2278–2282.
- [21] V. Pratap *et al.*, “Scaling speech technology to 1,000+ languages,” *Journal of Machine Learning Research*, vol. 25, no. 97, pp. 1–52, 2024.
- [22] W. N. Havard *et al.*, “Technologies de la parole et données de terrain : le cas du créole haïtien,” in *Actes de la 31ème Conférence sur le Traitement Automatique des Langues Naturelles, volume 1 : articles longs et prises de position*, M. Balaguer *et al.*, Eds. Toulouse, France: ATALA and AFPC, 7 2024, pp. 686–694.
- [23] D. Fattier, “Contribution à l’étude de la genèse d’un créole: l’atlas linguistique d’haïti, cartes et commentaires, 6 vol.” Bibliographical record, Presses Universitaires du Septentrion, Villeneuve d’Ascq, 1998, ph.D. Dissertation, Université de Provence.
- [24] FLA, Faculté de Linguistique Appliquée de l’Université d’Etat d’Haïti (anciennement Centre de Linguistique Appliquée (CLA)) *et al.*, “Atlas linguistique d’Haïti,” 2015. [Online]. Available: <https://cocoon.huma-num.fr/exist/crdo/meta/cocoon-8ea988d2-bf16-303d-81a0-0c55cc035240>
- [25] A. Valdman *et al.*, “On the influence of the standard norm of haitian creole on the cap haïtien dialect: Evidence from sociolinguistic variation in the third person singular pronoun,” *Journal of Pidgin and Creole Languages*, vol. 30, no. 1, p. 1–43, Apr. 2015.
- [26] J. Valk *et al.*, “VoxLingua107: a dataset for spoken language recognition,” in *Proc. IEEE SLT Workshop*, 2021.
- [27] V. Contributors, “Radio haiti collection,” 2024, digital items: 5,314; Total components: 3660; Last Indexed: 2024-12-05. [Online]. Available: <https://idn.duke.edu/ark:/87924/mlj07w>
- [28] H. Bredin *et al.*, “pyannote.audio: neural building blocks for speaker diarization,” in *ICASSP 2020*, Barcelona, Spain, May 2020.
- [29] —, “End-to-end speaker segmentation for overlap-aware ressegmentation,” in *Proc. Interspeech 2021*, Brno, Czech Republic, August 2021.
- [30] A. Conneau *et al.*, “Unsupervised cross-lingual representation learning for speech recognition,” in *Interspeech 2021*, 2021, pp. 2426–2430.
- [31] A. Paszke *et al.*, *PyTorch: an imperative style, high-performance deep learning library*. Red Hook, NY, USA: Curran Associates Inc., 2019.
- [32] M. Ott *et al.*, “fairseq: A fast, extensible toolkit for sequence modeling,” W. Ammar *et al.*, Eds. Minneapolis, Minnesota: Association for Computational Linguistics, Jun. 2019, pp. 48–53.
- [33] S. Evain *et al.*, “Task Agnostic and Task Specific Self-Supervised Learning from Speech with LeBenchmark,” in *Thirty-fifth Conference on NeurIPS (NeurIPS 2021)*, ser. NeurIPS 2021 Datasets and Benchmarks Track, on-line, United States, Dec. 2021. [Online]. Available: <https://hal.science/hal-03407172>
- [34] V. Pratap *et al.*, “Scaling speech technology to 1,000+ languages,” *Journal of Machine Learning Research*, vol. 25, no. 97, pp. 1–52, 2024.
- [35] A. Valdman, *Education Reform and the Instrumentalization of the Vernacular in Haiti*. Boston, MA: Springer US, 1982, pp. 139–170.
- [36] Unicode, “UAX #15: Unicode Normalization Forms,” 2024. [Online]. Available: <https://unicode.org/reports/tr15/>
- [37] J. Lehečka *et al.*, “Transfer learning of transformer-based speech recognition models from czech to slovak,” in *Text, Speech, and Dialogue*, K. Ekštejn *et al.*, Eds. Cham: Springer Nature, 2023, pp. 328–338.
- [38] M. Adda-Decker *et al.*, “Do speech recognizers prefer female speakers?” in *Interspeech 2005*, 2005, pp. 2205–2208.